CONTROLUNION	QUALITY & CONDITION ASSESSMENT SURVEY - STANDARD OPERATING PROCEDURE	Doc number	CUC/QP/016
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Fresh Apples — Quality Assessment at Destination

Covers both quality defects (e.g. undersize, misshape, sunburn) and condition defects (e.g. decay, mould, bruising, chilling damage)

1. Purpose

This Standard Operating Procedure (SOP) defines the method to be followed by Control Union inspectors when conducting a damage and/or quality assessment survey on a consignment of fresh apples following an allegation of quality deterioration, including decay, rot or other condition defects, reported at destination.

This SOP is issued in support of the Apple Inspection Manual – Destination Version 2 (APL-INS-MAN-DES-02) and shall be read together with the applicable Inspection Checklist – Apple Quality & Condition Assessment (CUC/FM/054).

The purpose of the survey is to establish and document the condition of the cargo as found at the time of inspection, based on representative sampling, objective observations, applicable measurements and photographic evidence.

2. Scope

This SOP applies to ad-hoc damage and quality surveys carried out at a client's, consignee's or storage facility in response to a complaint or allegation concerning the condition of a consignment of apples already at destination.

It applies to all Control Union inspectors and surveyors conducting the exercise on behalf of Control Union Kenya.

This SOP is intended for destination assessments and does not replace procedures applicable to routine loading, discharge, pre-shipment or other specialized inspections.

3. Roles and Responsibilities

Role	Responsibility
Requesting Party / Client	Raises the allegation, provides relevant shipment information and facility contact details, and facilitates site access.
Coordinator – Kenya Office	Confirms the appointment date, time and location; obtains available shipment information; notifies relevant stakeholders; and assigns the inspector.
Lead Inspector	Conducts the physical survey in accordance with this SOP and the Apple Inspection Manual; records observations and measurements; completes the Inspection Checklist; takes photographic evidence; and submits the draft report.
Reviewing Manager	Reviews the inspection findings, sampling records, defect calculations and tolerance scoring before the final report is issued.

4. Pre-Survey Preparation

Before attending the inspection, the assigned inspector shall, where applicable:

1. Confirm the inspection date, time, exact facility address and site contact name and telephone number.
2. Obtain available shipment identification and cargo information, including:
    - Packing list;
    - Container number(s);
    - Pallet numbers;

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- Box/packaging label information;
- Variety and category;
- Grower/packer information, where available;
- Packing date, where available; and
- Previous inspection certificates, reports or inspection stamps.

3. Prepare and verify the condition of the equipment available for the assignment, including:
    - Weighing scale (1 pc);
    - Calibration masses – 500 g and 250 g (2 pcs) – used to verify the weighing scale's accuracy immediately before use;
    - Probe thermometers (2 pcs);
    - Cutting machine (1 pc) – for internal examination of fruit where internal defects are suspected.
4. Confirm the availability of a refractometer and a penetrometer for the assignment. Where neither is available, this shall be recorded as an equipment limitation per Section 6.3 and Section 6.4 of this SOP, and the Coordinator shall attempt to source the missing instrument(s) before the inspection date.
5. Carry the applicable Inspection Checklist – Apple Quality & Condition Assessment (CUC/FM/054).
6. Carry a camera or suitable mobile device for photographic documentation.
7. Use PPE appropriate for the inspection environment, particularly where the cargo is stored in a cold room or warehouse, and where the cutting machine is used.
8. Review the relevant variety/group classification and tolerance requirements in the Apple Inspection Manual before commencing the inspection.

5. Sampling Procedure

Sampling shall follow the method specified in Section 2 of the Apple Inspection Manual.

5.1 Sample Selection

The inspector shall, wherever possible, select samples representing the distinct parameters within the consignment, including:

- Variety;
- Category;
- Grower;
- Size; and
- Packing date.

Where complete information is not available, sample selection shall be based on the information available from packing lists, pallet records, box labels or other shipment documentation.

5.2 Standard Sample

For a regular apple inspection:

- Select 6 boxes from 6 different pallets;
- Select 1 box per pallet; and

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- Take boxes from the 2nd row or below, and not from the top row.

Additional samples may be taken where the initial sample is considered insufficient to adequately represent the lot. Any additional sampling shall be clearly recorded on the Inspection Checklist, including the reason for the additional sampling.

5.3 Previously Opened or Inspected Boxes

The inspector shall not re-inspect a box that has already been opened or stamped by another quality control entity.

Where such boxes are encountered, the inspector shall record the relevant box/pallet reference and any visible inspection mark or stamp.

6. Instrument Readings

The following readings shall be taken and recorded in accordance with the Apple Inspection Manual and the equipment available for the assignment.

6.1 Pulp Temperature

Pulp temperature shall be measured using the probe thermometers. Where possible:

- Measure during container discharge; and
- Measure again in the cold room/storage area where the fruit is already in storage.

With two probe thermometers available, readings may be taken at both locations concurrently, or used to cross-check one another at a single location.

The Apple Inspection Manual identifies 0°C–1°C as the optimum travel temperature for apples.

The actual reading, location and time of measurement shall be recorded.

6.2 Weight / Undersize Verification

Fruit weight shall be measured using the weighing scale, checked against the size/weight tables for the relevant variety in the Apple Inspection Manual.

Before use, the weighing scale shall be verified using the calibration masses (500 g and 250 g). The verification result shall be recorded on the Inspection Checklist before any fruit is weighed. If the scale reading deviates materially from the known mass, the inspector shall not rely on the scale until it has been corrected or replaced.

6.3 Brix

Brix (°Bx) measurement requires a refractometer. Where a refractometer is available for the assignment, one piece of normal-looking fruit shall be used for the initial reading; where a reading appears unusual or inconsistent, a fresh fruit shall be tested and the result recorded.

Where no refractometer is available, Brix shall not be tested. This shall be recorded as an equipment limitation under Section 10 of this SOP and on the Inspection Checklist, rather than omitted without explanation.

6.4 Firmness

Firmness measurement requires a penetrometer. Where a penetrometer is available for the assignment:

- Test each fruit twice;
- Take readings on opposite sides of the equatorial section;

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- Peel the skin at the test points; and
- Avoid bruised or otherwise damaged areas when selecting the test points.

The Apple Inspection Manual identifies 14 lb as the minimum expected arrival firmness.

Where no penetrometer is available, instrumented firmness testing shall not be performed. Given that softness is a condition defect directly relevant to allegations of quality deterioration, the inspector shall still record a qualitative, non-instrumented assessment of firmness by hand for each sample, clearly labelled as a tactile observation and not an instrument reading. The absence of a penetrometer shall be recorded as an equipment limitation under Section 10 of this SOP and on the Inspection Checklist. The Coordinator shall attempt to source a penetrometer ahead of the inspection date wherever possible.

6.5 Internal Examination

Where internal decay, core flush, internal breakdown or another suspected internal defect requires the fruit to be cut, the cutting machine shall be used to obtain a clean, consistent cut suitable for examination and photographic documentation.

All readings and equipment verification results shall be recorded on the Inspection Checklist.

7. Physical Assessment of Defects

Each sampled box shall be examined systematically, and the presence and severity of relevant defects shall be recorded.

Defect definitions and visual references contained in the Apple Inspection Manual shall be applied consistently.

7.1 Quality Defects

The following quality defects shall be assessed, where applicable:

- Undersize;
- Under colour;
- Skin defects;
- Russet/scars – slight/moderate and severe;
- Foreign material/dirty;
- Mechanical damage;
- Peduncle broken/absent;
- Misshape; and
- Sunburn.

7.2 Condition Defects

Particular attention shall be given to condition defects because these may develop or worsen during storage and are central to an allegation of quality deterioration.

The following shall be assessed, where applicable:

- Decay;
- Mould;
- Softness;
- Bruising;

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- Bitter pit;
- Scald;
- Dry fruit/shrivelling;
- Watercore damage;
- Core flush;
- Chilling/freezing damage;
- Internal breakdown;
- Open wounds; and
- Insect presence/damage.

7.3 Decay

Where decay is suspected, the inspector shall carefully examine the affected fruit.

The Apple Inspection Manual describes decay as potentially involving brown colouration, disintegration, aqueous flesh and a fetid/fermentation odour. In some cases, decay may be internal and may only be identified after cutting the fruit.

Where internal decay or another internal defect is suspected, representative fruit shall be cut and the findings documented photographically.

7.4 Defect Recording

For each sample, the inspector shall record:

- Number of fruits examined;
- Number of fruits affected;
- Defect type;
- Severity/category, where applicable;
- Percentage affected; and
- Relevant observations.

Where a fruit exhibits more than one defect, the inspector shall record the defects in accordance with the applicable requirements of the Apple Inspection Manual and Inspection Checklist.

8. Scoring Against Tolerances

The inspector shall tally the percentage of fruit affected by quality defects and condition defects separately and assess the results against the applicable QC tolerance requirements in the Apple Inspection Manual.

The following scoring framework shall be applied where applicable:

QC Score	Total Quality Defects	Total Condition Defects
1 – Very Good	> 0 – 5%	> 0 – 1%
2 – Good	> 5 – 10%	> 1 – 10%
3 – Fair	> 10 – 20%	> 10 – 20%
4 – Poor	> 20 – 30%	> 20 – 30%
5 – Very Poor	> 30%	> 30%

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Individual defect percentages shall also be recorded for every quality and condition defect listed in Sections 7.1 and 7.2 of this SOP (including, but not limited to, Decay, Mould, Bruising, Bitter Pit, Scald, Watercore Damage and Internal Breakdown).

The final assessment shall clearly distinguish between quality defects and condition defects.

Where the Apple Inspection Manual provides a specific tolerance for an individual defect, that specific requirement shall be applied.

9. Photographic Evidence

Photographic evidence shall be sufficient to support the findings recorded in the Inspection Checklist and survey report.

The inspector shall, where applicable, photograph:

1. Exterior condition of the container, warehouse or storage facility;
2. General storage conditions;
3. Pallets and pallet identification;
4. Box labels and shipment identification;
5. Sampled boxes before opening;
6. Any existing inspection stamps or markings;
7. Representative examples of each significant defect identified;
8. Affected fruit showing the extent and severity of damage; and
9. Cut fruit where internal defects such as decay, core flush or internal breakdown are suspected.

Photographs shall be identifiable and cross-referenced to the relevant sample, box or pallet reference wherever practicable.

The inspector shall avoid taking photographs that cannot be linked to the sampled cargo or that do not provide meaningful evidentiary value.

10. Evidence and Condition at Time of Inspection

The inspection shall establish and document the condition of the cargo as observed at the time of inspection.

The inspector shall record, where applicable:

- Whether the cargo had been moved, sorted, repacked or otherwise handled before inspection;
- Whether the cargo had previously been inspected;
- Existing inspection stamps or markings;
- Storage duration, where known;
- Storage conditions;
- Temperature conditions;
- Any relevant information provided by the consignee or facility representative; and
- Any limitations encountered during the inspection, including any required instrument (e.g. refractometer or penetrometer) that was not available for the assignment.

The inspector shall distinguish clearly between:

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- Observed facts;
- Measurements and calculated results;
- Information provided by third parties; and
- Inspector's professional observations.

The inspector shall not attribute the cause or responsibility for the damage to the shipper, carrier, consignee, storage facility or any other party unless such attribution is supported by objective evidence and falls within the agreed scope of the assignment.

Where the cause cannot be objectively established, the report shall state the observed condition without assigning causation.

11. Reporting

1. Complete the Inspection Checklist – Apple Quality & Condition Assessment (CUC/FM/054) in full, including sample identification, instrument readings, defect observations and calculations.
2. Prepare a draft survey report summarising:
    - Assignment details;
    - Cargo identification;
    - Sampling methodology;
    - Number of boxes/fruits examined;
    - Instrument readings;
    - Defects identified;
    - Individual defect percentages;
    - Quality and condition defect totals;
    - Applicable QC score(s);
    - Relevant observations and limitations; and
    - Supporting photographic evidence.
3. Submit the draft report and completed checklist to the Reviewing Manager for quality review.
4. Address any review comments or required corrections.
5. Issue the final report to the client/requesting party within the agreed turnaround time.

The report shall be factual, objective and traceable to the inspection records and photographic evidence.

12. Records Retention

The following records shall be retained in the applicable job file in accordance with Control Union's document retention requirements:

- Inspection Checklist;
- Survey report;
- Photographs;
- Instrument readings;
- Sampling records;

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- Relevant shipment documentation;
- Correspondence or instructions relevant to the assignment; and
- Any other supporting evidence generated or received during the inspection.

13. Health, Safety and Operational Considerations

The inspector shall comply with applicable site safety requirements and use appropriate PPE.

Attention shall be given to:

- Cold-room conditions;
- Safe access to pallets and stacked cargo;
- Manual handling;
- Slippery floors or wet areas;
- Safe operation of the cutting machine, including keeping hands clear of the blade and using any guard provided;
- Safe use of other inspection tools; and
- Any site-specific safety requirements communicated by the facility.

The inspector shall not undertake any activity that presents an unreasonable safety risk and shall report any significant safety concern to the responsible site representative and Control Union management.

14. Related Documents

The following documents shall be read and used together with this SOP, where applicable:

1. Apple Inspection Manual – Destination Version 2 — Document Reference: APL-INS-MAN-DES-02
2. Inspection Checklist – Apple Quality & Condition Assessment — Document Reference: CUC/FM/054

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